class starts
@ 9:05 pm

BASICS OF GenAI

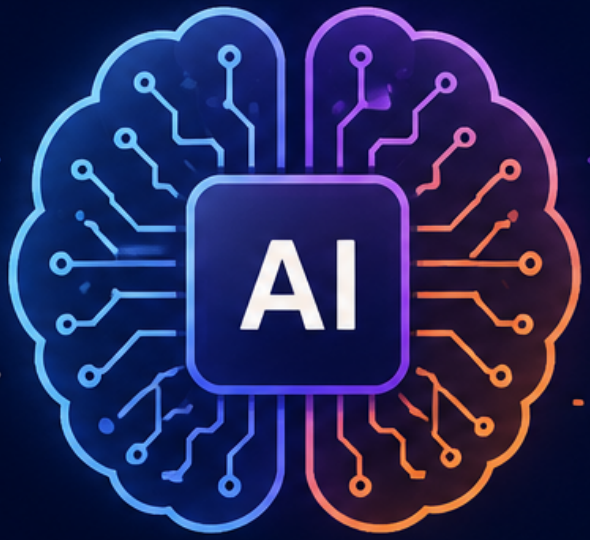
Understand. Create. Innovate.



TEXT GENERATION



IMAGE GENERATION

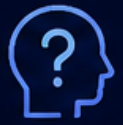


CODE GENERATION



IDEAS TO IMPACT





UNDERSTAND



CREATE



INNOVATE



EMPOWER

Lecture: Introduction to AI Engineering

Agenda

1. Introduction to AIGenAIneering
2. AI Engineering Ecosystem
3. LLM APIs & SDKs
4. Prompt Engineering
5. Open Source LLMs
6. Retrieval-Augmented Generation (RAG)
7. Vector Databases & Embeddings
8. LLM Evaluation

Recap

NLP → Attention Mechanism, Transformer, BERT

↳ Language Model

GPT



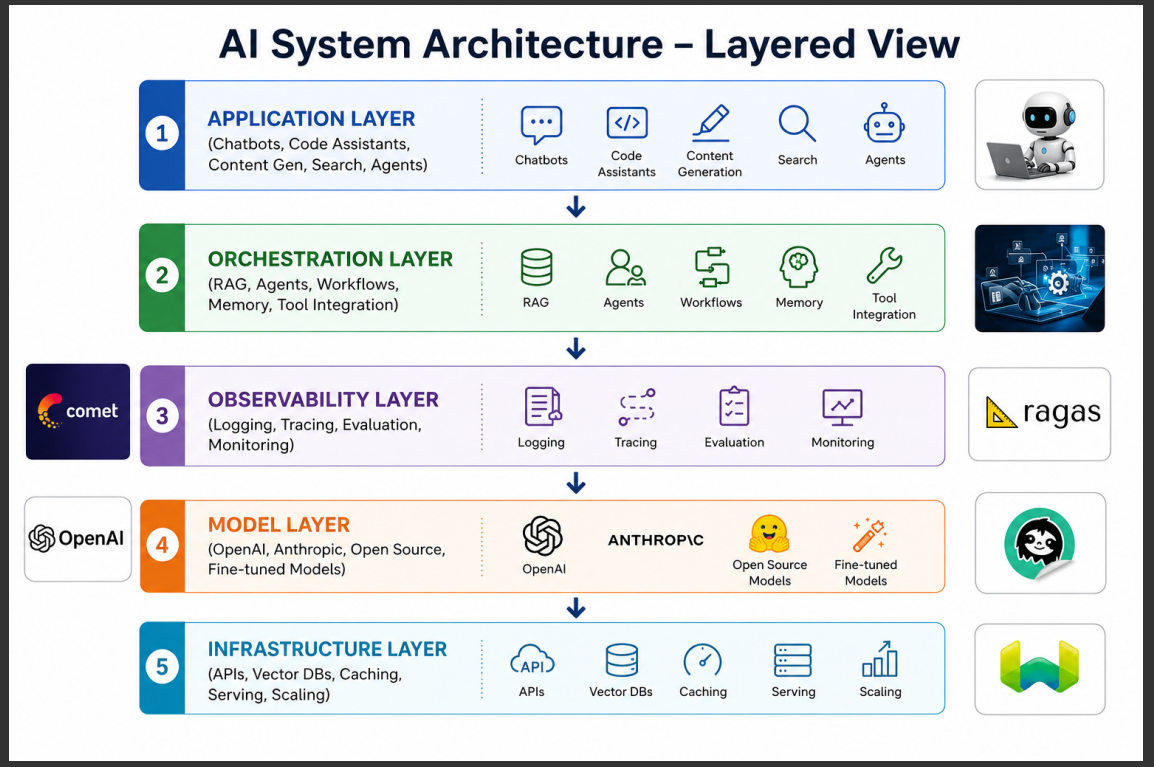
LLM

↳ Trainable Parameter

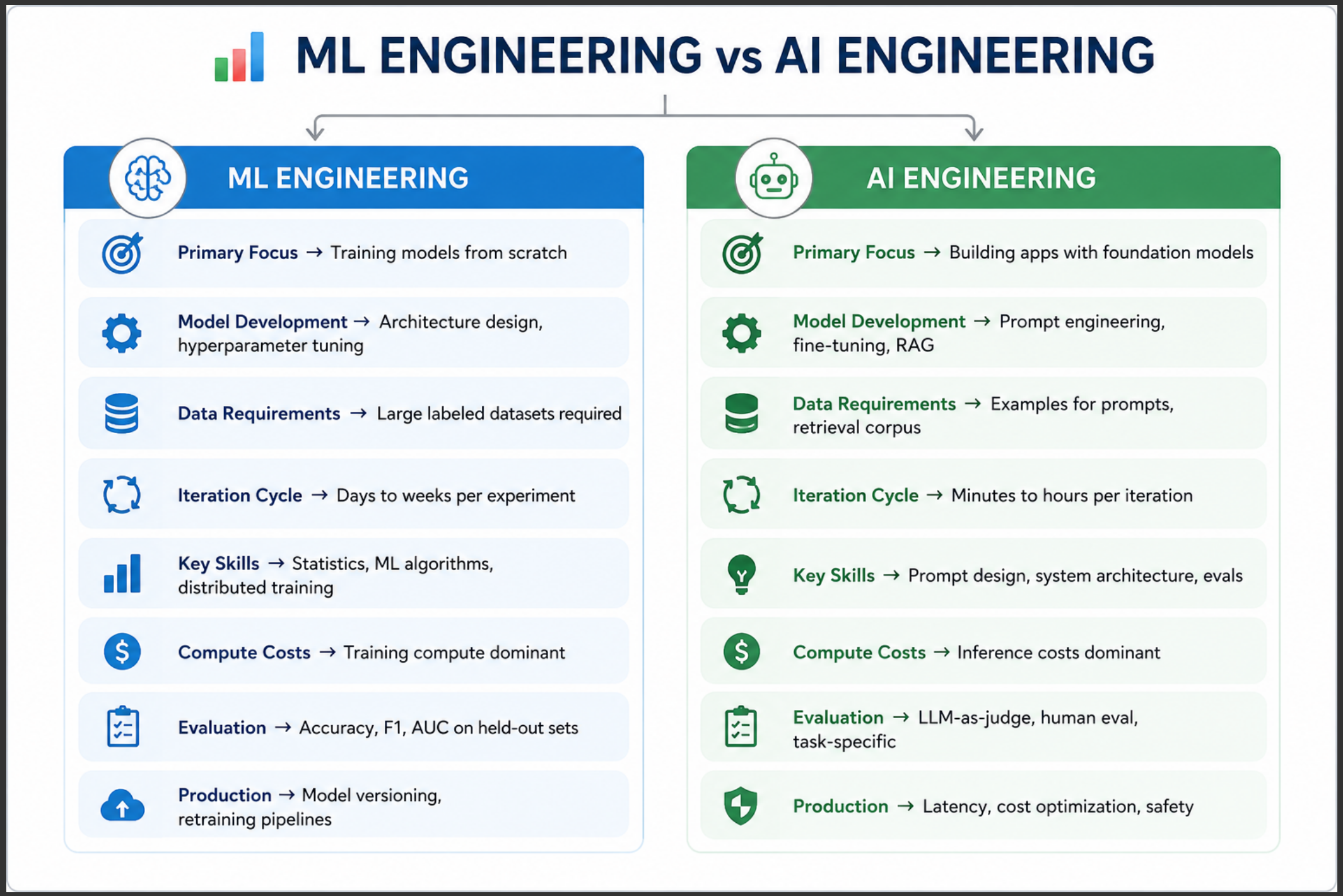
Definition -

AI Engineering is the discipline of building production applications powered by foundation models (LLMs, diffusion models, multimodal models). Unlike traditional ML engineering which focuses on training models from scratch, AI Engineering emphasizes:

- Integration of pre-trained foundation models
- Prompt design and optimization
- Orchestration of AI components (RAG, agents, tools)
- Evaluation and observability of AI systems
- Finetuning a Pre-trained Model
- Production concerns: latency, cost, reliability, safety



AI Engineering vs ML Engineering



1 LLM PROVIDERS

APIs for accessing language models

Private

- OpenAI
- Google
- Anthropic

Open-source

- OpenRouter
- Hugging Face
- Together AI

2 LLM APPLICATION DEVELOPMENT ECOSYSTEM

Category	Tools
Vector Databases	Chroma, Pinecone, Milvus, FAISS, Weaviate
Finetuning	Huggingface TRL
Quantization	llama.cpp (GGUF), GPTQ, AWQ, SmoothQuant
Evaluation	Ragas, DeepEval, LM Evaluation Harness
Security	garak (vulnerability testing), langfuse (monitoring)
Experiment Tracking	Weights & Biases, Comet, MLflow

3 LLM FRAMEWORKS

Category	Frameworks
Orchestration	LangChain, LlamaIndex, MCP
Agents	LangGraph, smolagents, CrewAI, AutoGen
Fine-Tuning	TRL, Unsloth, Axolotl
Structured Output	Outlines, LMQL
Programmatic LLMs	DSPy

4 LLM INFRASTRUCTURE

Deployment Type	Tools
Local	LM Studio, Ollama, kobold.cpp
Demo/Prototype	Gradio, Streamlit, HF Spaces
Production Server	vLLM, TGI, Ray Serve
Edge	MLC LLM, mnn-llm
Cloud Orchestration	SkyPilot

5 MODEL HUBS & DATA

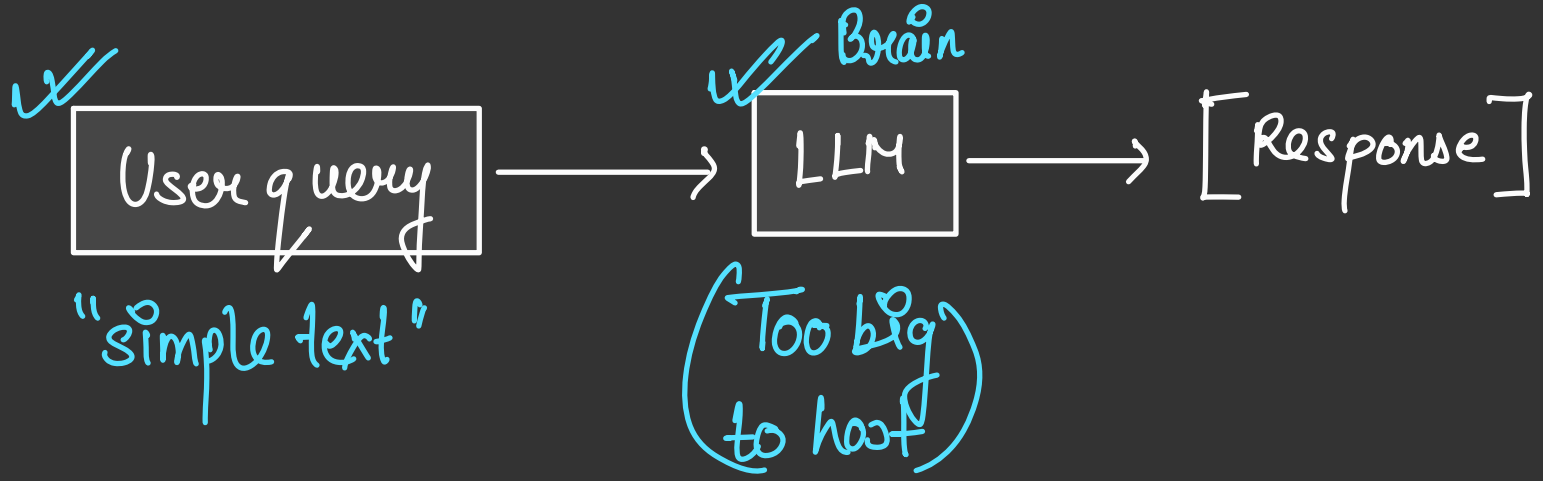
Model Hubs	Hugging Face Hub, Ollama Library
Data Tools	NeMo-Curator, Distilabel
Embedding Models	Sentence Transformers, MTEB Leaderboard

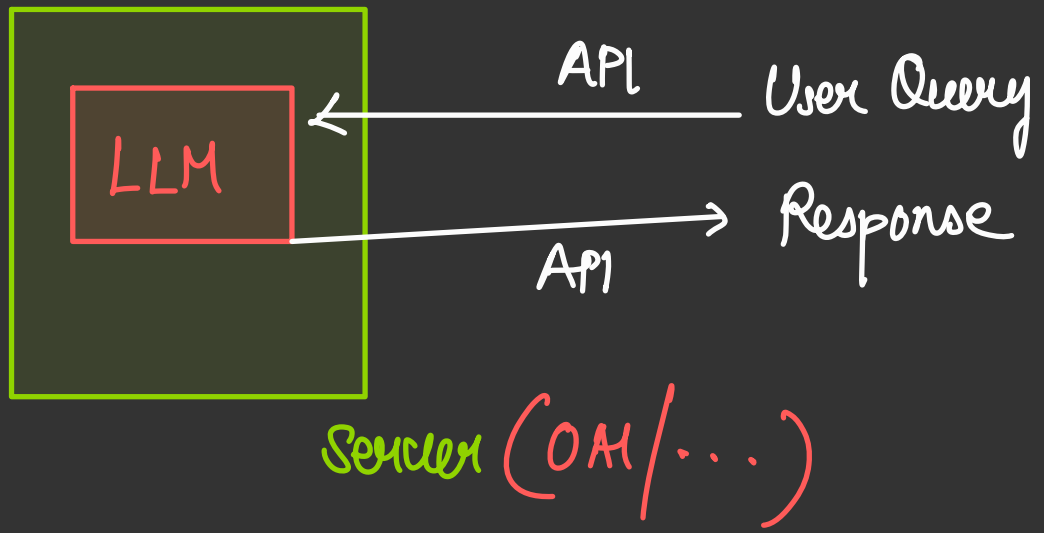
6 QUICK REFERENCE BY USE CASE

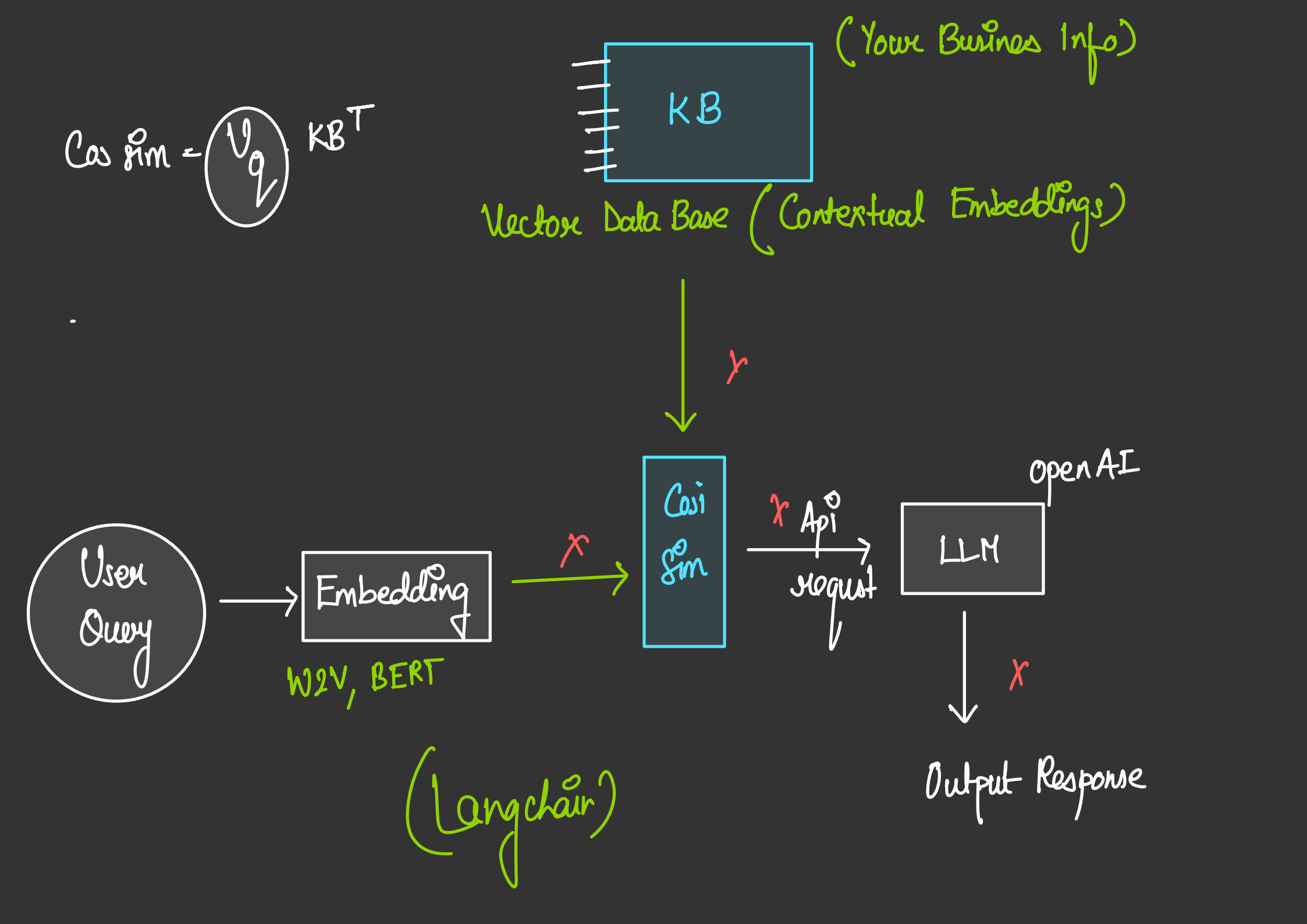
Goal	Stack
Build RAG app	LangChain/LlamaIndex + Vector DB + Embedding Model
Fine-tune model	TRL/Unsloth + HF Transformers + W&B
Deploy locally	Ollama/LM Studio + GGUF models
Production inference	vLLM/TGI + Cloud/On-prem GPUs
Build agents	LangGraph/smolagents + Tool integrations

AI Chatbots

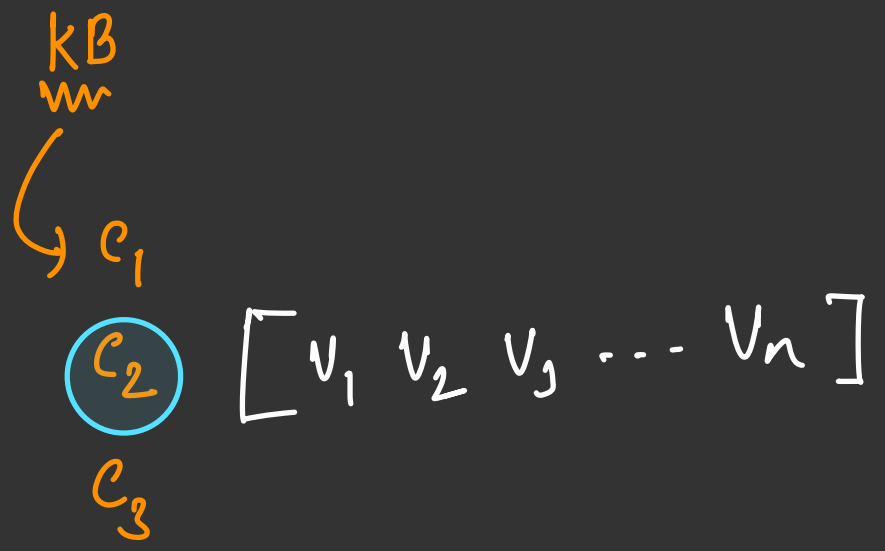
google / Anthropic / OAI



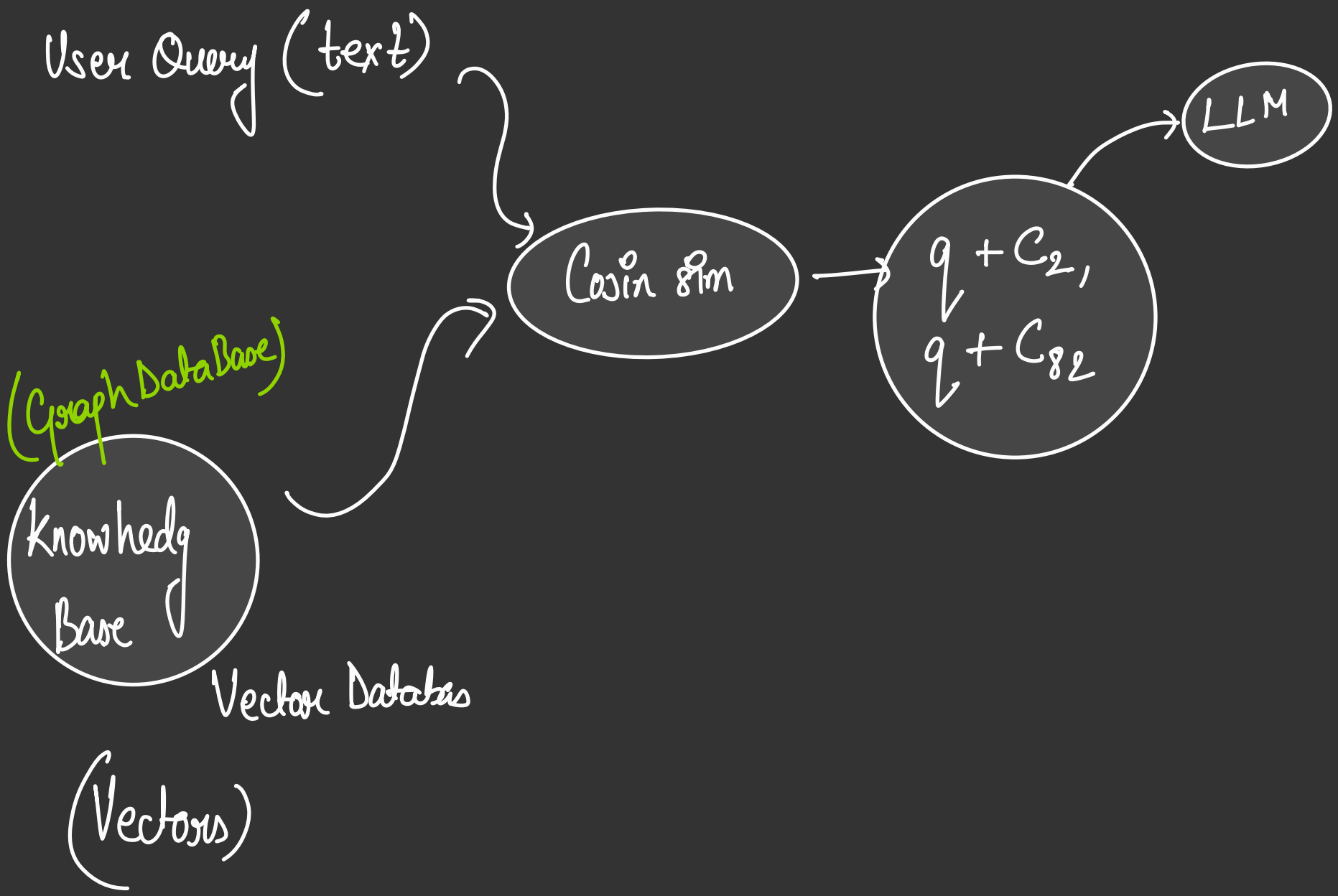



(Your Busines Info)
KB
Cos sim = V_q KB^T
Vector Data Base (Contextual Embeddings)
x
User Query
Embedding
W2V, BERT
x
Cosi Sim
x Api request
LLM
open AI
x
Output Response
(Langchain)

q = Tell me where is my order?



q + c_2


User Query (text)
Cosin Sim
q + C2,
q + C82
LLM
(Graph DataBase)
Knowledg Base
Vector Database
(Vectors)

OpenAI API vs. HF Transformers (Local)

Aspect	OpenAI API	HF Transformers (Local)
Setup	API key only	Download model (~GB)
Cost	Pay per token	Free (GPU costs)
Latency	Low	Depends on hardware
Privacy	Data sent to OpenAI	Fully local
Models	GPT-4, GPT-4o, etc.	500k+ open models

What is RAG?

Retrieval-Augmented Generation

RAG is an AI framework that combines the power of retrieval from external knowledge sources with the generative capabilities of LLMs to produce accurate, relevant, and up-to-date answers.

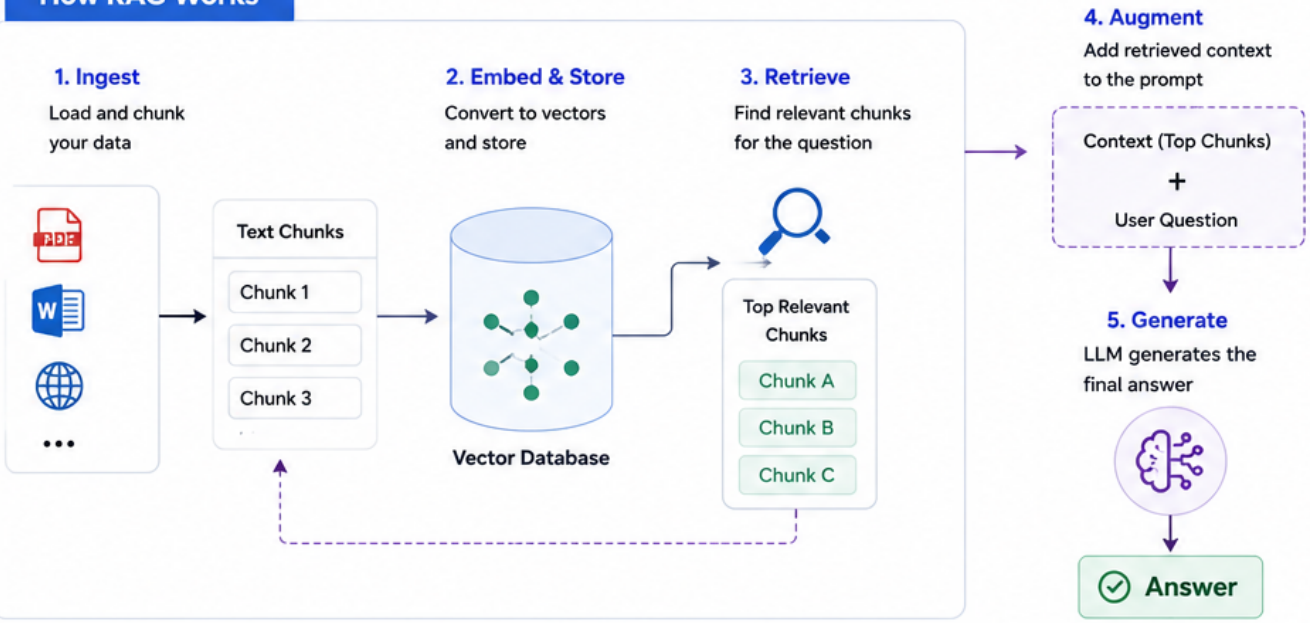
Why RAG?

- **More Accurate** — Grounds answers in trusted sources
- **Up-to-date** — Uses your latest data
- **Reduce Hallucinations** — LLM answers are fact-checked by context
- **Cost Efficient** — No need to fine-tune for new knowledge

Key Components

- **Knowledge Base** — Your private data (docs, PDFs, websites, DBs, etc.)
- **Embeddings** — Convert text into vectors to capture meaning
- **Retriever** — Find the most relevant information from the knowledge base
- **LLM (Generator)** — Generate the final answer using retrieved context and the user question

How RAG Works



Example

User Question
What is the refund policy?

RAG Answer
According to the refund policy document, customers can request a refund within 30 days of purchase...

Common Use Cases

- Customer Support
- Knowledge Assistants
- Research & Summarization
- Legal & Compliance
- Healthcare

Bottom Line: RAG helps LLMs give reliable, contextual, and up-to-date answers by looking up facts before generating a response.

prompt

Frame this Answer

User query : { query } → placeholder

Answer : { context vector/chunk(s) }



much more exhaisive

Topics

1) Introduction
2) RAG, Model Arch, Ragas, Comet ML
3) LLM API (open AI)
4) Design AI Apps (Eval pipeline & Observability)
5) prompt Engg
6) RAG
7) Embedding deep dive
8) Multi modal & tabular RAG
9) Security

#Advance

1) Agents: Foundations & Planning
2) Agent concepts
3) Frameworks & protocol (MCP)
4) Fine Tuning
5) AI Modalities
6) Scaling & AI application
7) Model quantization techniques
8) Advanced Fine tuning.
9) Inference optimization

ANN (Approx Nearest Neighbour) → Qdrant

IVF (Inverted File Index) → FAISS (meta)

Why Vector Databases?

Traditional databases search by **exact match** (WHERE name = 'John').
Vector databases search by **similarity** (find items most similar to this embedding).
This enables:

Semantic search
"affordable laptops" matches "budget notebooks"

RAG pipelines
Retrieve relevant context for LLM generation

Recommendation systems
Find similar products, content, users

Anomaly detection
Identify outliers in embedding space

How They Work

1 INDEXING



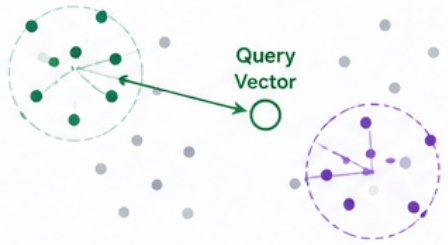
2 QUERYING



Distance Metrics

Metric	Formula	Best For
Cosine Similarity	$\cos(\theta) = \frac{A \cdot B}{\|A\|\|B\|}$	Text embeddings (normalized)
Euclidean (L2)	$\sqrt{\sum_{i=1}^{n} (a_i - b_i)^2}$	Image embeddings
Dot Product	$\sum_{i=1}^{n} (a_i \times b_i)$	When magnitude matters

Visual Intuition



Points closer to the query vector are more similar.

Comet ML: Complete LLM Tracking Pipeline

Track. Compare. Optimize. Ship with Confidence.

What is Comet ML?

Comet ML is an **experiment tracking** and **observability platform** for machine learning. It helps data science and LLM teams track prompts, fine-tuning experiments, model evaluations, and production performance — all in one place.

Why Comet for LLMs?

- **End-to-end visibility from prompts to production**
  Track every experiment, evaluation, and deployment.
- **Compare & reproduce with confidence**
  Side-by-side comparisons and reproducible results.
- **Monitor quality in production**
  Detect regressions and track real-world performance.
- **Built for collaboration**
  Share insights, annotate runs, and align your team.

The Complete LLM Tracking Pipeline

1. **Prompt Tracking**
   - Log prompts, variables and versions
   - Track prompt templates and changes
2. **Dataset Logging**
   - Log datasets and data versions
   - Track preprocessing and splits
3. **Fine-tuning Experiments**
   - Track hyperparameters
   - Log metrics, losses and checkpoints
4. **Evaluation**
   - Evaluate with RAGAS, custom metrics
   - Track quality over experiments
5. **Model Registry**
   - Version and register the best models
   - Add tags, notes and metadata
6. **Deployment Tracking**
   - Track deployments and environments
   - Monitor usage and latency
7. **Production Monitoring**
   - Monitor drift, quality and feedback
   - Set alerts for regressions

What You Can Track

Prompts & Responses | Datasets | Hyperparameters & Metrics | Evaluations | Models & Versions | Deployments | Production Metrics

Key Benefits

- **Improve model quality**
  Identify what works and what doesn't
- **Save time & compute**
  Reproduce and compare experiments easily
- **Collaborate better**
  Share experiments and insights
- **Ship with confidence**
  Monitor and maintain model performance

Comet Dashboard Preview

